

ARYAN KUMAR SINHA

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Summary

My name is Aryan. a Computer Science graduate pursuing M.Tech in AI & ML with hands-on experience in machine learning, computer vision, NLP, and backend development. Skilled in building production-oriented applications and research-driven ML solutions. Passionate about building practical software systems and solving real-world problems through scalable and data-driven solutions.

EXPERIENCE

La Trobe University

Oct. 2023 – Apr. 2024

Research Intern

- Developed a machine learning-based framework to detect smishing attacks, achieving **98% accuracy** using ANN models
- Collected and processed **50,000+ SMS datasets**; applied feature engineering techniques including TF-IDF and word embeddings
- Implemented and optimized multiple models, improving detection rates by **50%** and reducing analysis time by **30%**
- Built a user-friendly mobile application for real-time smishing detection, increasing detection speed by **40%**
- Gained practical experience in machine learning, data processing, and cybersecurity through real-world application

PROJECTS

Multi-Modal Earnings Call Volatility Prediction

2026

- Built a multi-modal pipeline combining earnings call transcripts, audio sentiment, and historical stock data for volatility prediction
- Processed and analyzed **500+ earnings calls** using NLP techniques, audio feature extraction, and financial time-series preprocessing
- Trained and evaluated multiple machine learning models including Praat, FinBERT, wav2vec2, and ensemble methods to compare forecasting performance
- Gained practical experience in NLP, audio processing, feature engineering, and financial machine learning workflows

ScannerontheGO

2025

- Engineered a Flask-based computer vision pipeline for stable frame extraction and automated PDF generation from videos
- Developed an automated frame extraction algorithm achieving **99% accuracy** in detecting non-blurry frames
- Implemented asynchronous processing and optimized the document generation pipeline, reducing processing latency by **35%**
- Developed modular backend APIs for video upload, frame processing, and document generation workflows enabling scalable future integration

Setting High Score in Subway Surfer

2025

- Developed a DDQN reinforcement learning agent for autonomous Subway Surfer gameplay optimization
- Built a custom training environment with TensorFlow Extended (TFX) pipeline for training, evaluation and inference workflow
- Achieved a high score of **14,569,684**, placing within the top **1%** of global players through continuous policy optimization

EDUCATION

Master of Technology in Artificial Intelligence & Machine Learning

Aug. 2025 – May 2027

Symbiosis Institute of Technology, Pune, MH

GPA: 7.15

Bachelor of Technology in Computer Science & Engineering

Aug. 2020 – May 2024

Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology, Chennai, TN

CGPA: 8.57

CERTIFICATIONS

Data Analysis for Biologists | NPTEL

2023

CCNAv7: Introduction to Networks | Cisco

2022

Cybersecurity Essentials | Cisco

2022

Architecture With VMware NSX | Coursera

2020

SKILLS

Languages: Java, Python, C, SQL, Bash

Frameworks & Libraries: JavaFx, Spring Boot, Flask, cProfile, Pygame, Tkinter, OpenCV, SAM, TensorFlow, Scikit-learn, Pandas, Selenium, Playwright

Web & APIs: REST APIs, HTML, CSS, JavaScript

Databases: MySQL, MongoDB, ScyllaDB

AI/ML: Machine Learning, Deep Learning, NLP, Computer Vision, LLMs, Agentic AI, RAG, Feature Engineering

Tools & Platforms: Git, GitHub, Docker, Linux, Postman, VS Code, Power BI

Soft Skills: Communication, Team Collaboration, Critical Thinking, Adaptability, Time Management, Quick Learner